FORECASTING FUTURE LANGUAGE: CONTEXT DESIGN FOR MENTION MARKETS

Sumin Kim[1,*], **Jihoon Kwon**[1,*], **Yoon Kim**[2], **Nicole Kagan**[3],
Raffi Khatchadourian[4], **Wonbin Ahn**[5], **Alejandro Lopez-Lira**[6],
Jaewon Lee[7], **Yoontae Hwang**[8], **Oscar Levy**[9], **Yongjae Lee**[10,**], **Chanyeol Choi**[1,**]

[1]LinqAlpha [2]Massachusetts Institute of Technology [3]Kalshi [4]IBM [5]LG AI Research
[6]University of Florida [7]Seoul National University [8]Pusan National University
[9]University of California, Berkeley [10]UNIST

[*]Equal Contribution. [**]Corresponding author.

arXiv:2602.21229v2 [q-fin.GN] 27 Feb 2026

ABSTRACT

Mention markets, a type of prediction market in which contracts resolve based on whether a specified keyword is mentioned during a future public event, require accurate probabilistic forecasts of keyword-mention outcomes. While recent work shows that large language models (LLMs) can generate forecasts competitive with human forecasters, it remains unclear how input context should be designed to support accurate prediction. In this paper, we study this question through experiments on earnings-call mention markets, which require forecasting whether a company will mention a specified keyword during its upcoming call. We run controlled comparisons varying (i) which contextual information is provided (news and/or prior earnings-call transcripts) and (ii) how *market probability*, (i.e., prediction market contract price) is used. We introduce Market-Conditioned Prompting (MCP), which explicitly treats the market-implied probability as a prior and instructs the LLM to update this prior using textual evidence, rather than re-predicting the base rate from scratch. In our experiments, We find three insights: (1) richer context consistently improves forecasting performance; (2) market-conditioned prompting (MCP), which treats the market probability as a prior and updates it using textual evidence, yields better-calibrated forecasts; and (3) a mixture of the market probability and MCP (MixMCP) outperforms the market baseline. By dampening the LLM's posterior update with the market prior, MixMCP yields more robust predictions than either the market or the LLM alone.

1 INTRODUCTION

Prediction markets – markets in which participants trade contracts on future events – use contracts that pay a fixed payoff depending on the realized outcome. As a result, contract prices are commonly interpreted as the probability of the outcome as collectively assessed by market participants (Wolfers & Zitzewitz, 2004; Arrow et al., 2008). For example, a contract that pays $1 if global oil prices rise by more than 10% over the next quarter may trade at $0.55, implying a 55% market-implied probability. Can probability assessments produced by prediction markets be improved by incorporating information from text, such as news and corporate disclosures, using modern NLP methods?

Recent work shows that large language models (LLMs) can serve as probabilistic forecasters for open-domain event questions, with probability estimates that are comparable to prediction-market forecasts (Halawi et al., 2024; Alur et al., 2025). This naturally raises the question of whether LLMs and prediction markets can be complementary, with LLM-based signals helping to refine market-implied probabilities. However, it remains unclear how LLM-based signals can be used to improve the predictive accuracy of market-implied probabilities, rather than competing with or replacing market forecasts.

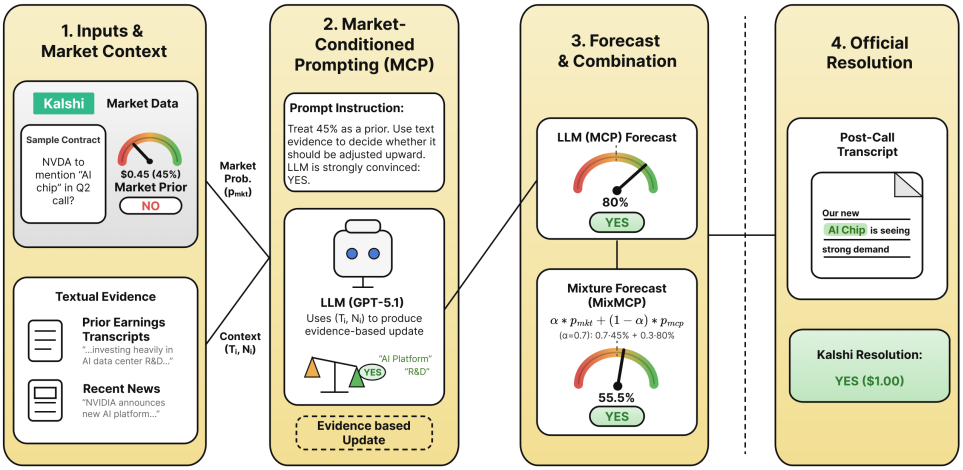

Figure 1: **Resolving Market Ambiguity with MIXMCP.** The diagram illustrates a mid-confidence scenario where the *market probability* is uncertain (45%). By conditioning on relevant textual evidence (e.g., earnings transcripts, news), MCP identifies a strong signal ($p_{\text{mcp}} = 80\%$) that the market has not fully priced in. Finally, MIXMCP combines the stable market prior with the LLM's evidence-based update ($\alpha = 0.7$), producing a refined forecast (55.5%) that correctly anticipates the resolution.

In this paper, we first examine whether LLM-based signals can help improve market-implied forecasts in a specific class of prediction markets, rather than competing with them. Specifically, we study mention markets—a class of prediction markets that resolve on whether a specified keyword appears verbatim in an upcoming earnings-call transcript. Mention markets provide a particularly suitable testbed because forecasting their outcomes amounts to predicting future language, a task well aligned with LLMs' language understanding capabilities, and because language used in earnings calls is closely linked to economically meaningful outcomes such as stock price reactions (Price et al., 2012; Cicon, 2017).

Our proposed Market-Conditioned Prompting (MCP) method treats the market-implied probability(*market probability*) as a prior and uses the model to update this prior based on external textual context. By explicitly conditioning the model on the market probability, MCP elicits evidence-based revisions rather than independent forecasts.

Our contributions can be summarized as:

- **Framework:** We formalize text-grounded forecasting as a *market-conditioned updating* problem, where LLMs use textual context as evidence to revise the *market probability*, rather than acting as standalone predictors.
- **Methodology:** We propose MCP, a prompting protocol that explicitly conditions LLMs on market prices, leading to significantly better calibration than standard prompting approaches.
- **Performance:** We show that a conservative mixture of the market prior and the MCP posterior (MixMCP) consistently outperforms the market baseline, confirming that LLMs can add value even to efficient markets.

2 BACKGROUND

Earnings-Call Mention Markets. Kalshi operates prediction markets on earnings-call *mentions*, where each contract resolves to `YES` if a specified keyword appears verbatim in the official post-call transcript. A market instance i is defined by an earnings event e_i, a target keyword k_i, and a pre-call cutoff time t_i. At t_i, the YES-contract price reflects the *market probability* p_i^{MKT} that the keyword will be mentioned. As a text-grounded form of prediction market, earnings-call mention markets retain the standard interpretation of contract prices as implied probabilities, while shifting the prediction target to future language. Because settlement is determined by exact string matching against the official

post-call transcript, outcomes are objectively verifiable and evaluation is unambiguous. Keywords are externally specified through Kalshi's market-creation process, yielding precisely defined contracts and preventing researcher intervention in keyword selection. This process also enforces standard safeguards against manipulation and ensures ethical, well-posed market design.

Forecasting Task. Given information available at t_i—including the market price and pre-call textual evidence such as news and prior earnings-call transcripts—a forecaster outputs a probability $p_i \in [0, 1]$ for the binary outcome $y_i \in \{0, 1\}$ determined by Kalshi's official resolution.

Evaluation Metrics. Probabilistic forecasts are commonly evaluated using the Brier score (Brier, 1950; Gneiting & Raftery, 2007),

$$\text{Brier} = \frac{1}{N} \sum_{i=1}^{N} (p_i - y_i)^2 \,, \tag{1}$$

which measures the mean squared error between predicted probabilities and outcomes – lower is better. Calibration is assessed via Expected Calibration Error (ECE) (Guo et al., 2017). A well-calibrated forecaster should have actual accuracy match predicted confidence – e.g., among predictions made with 70% confidence, roughly 70% should resolve to `YES`. ECE quantifies this gap by partitioning predictions into B equal-width bins $S_b = \{i : p_i \in (\frac{b-1}{B}, \frac{b}{B}]\}$ and computing:

$$\text{ECE} = \sum_{b=1}^{B} \frac{|S_b|}{N} \, |\bar{y}_b - \bar{p}_b| \,, \tag{2}$$

where $\bar{y}_b$ and $\bar{p}_b$ are the mean outcome and mean predicted probability in bin b, respectively. Lower ECE indicates better calibration. We use B=10 and also report accuracy and F1 score Chinchor (1992).

3 METHODS

We study how to combine a strong market forecast with text based evidence available before the earnings call. The goal is to improve probabilistic accuracy without replacing the market prior.

Inputs and Base Forecaster. For each market instance i, we take the *market probability* at forecast time p_i^{mkt} and a prompt context c_i built from available text. A fixed language model with parameters θ maps the context to a probability $p_i^{\text{llm}} = f_\theta(c_i)$. We hold the model and decoding constant across all conditions.

Textual Context. We build two source of pre-call evidence. The transcript block T_i contains the prior quarter earnings call transcript for the same firm. The news block N_i contains retrieved company related news articles (Lewis et al., 2020) available before the cutoff t_i. We isolate the effect of each source by varying the context (Liu et al., 2023)

$$c_i \in \{\emptyset, N_i, T_i, (T_i, N_i)\}. \tag{3}$$

Market Signal as Plain Text. A direct baseline inserts the market probability into the prompt without asking the model to treat it as a prior. Let $\phi(\cdot)$ be a deterministic textual rendering of the market probability. The baseline forecast is

$$p_i^{\text{llm}} = f_\theta(T_i, N_i, \phi(p_i^{\text{mkt}})). \tag{4}$$

This condition tests whether the model improves by simple exposure to the market number alone.

Market-Conditioned Prompting (MCP). We propose Market-Conditioned Prompting (MCP), a prompting protocol in which a large language model is explicitly provided with the *market probability* together with pre-call textual evidence, and asked to output a revised probability forecast.